
Pop Stars Have Something to Teach CU Professors: How Low Can You Go with Sentence Embeddings to Stay Useful?

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Abstract

How much embedding dimensionality is needed for a given corpus? Can a model trained on a simpler domain transfer effectively to semantically richer tasks? How does sensitivity to embedding dimensionality vary across corpora?

To answer these questions, we study how domain and embedding dimensionality interact in unsupervised sentence encoders. Using two extreme corpora, **Taylor Swift lyrics** and **Nakul Verma’s papers**, we train **contrastive BERT models from 4D to 767D** and evaluate geometry and transfer learning properties.

Surprisingly, lower-dimensional **models trained on lyrics (4-256D) outperform paper-trained models on STS-B**, with papers only catching up at 767D. This suggests that simple, high-frequency text can be an efficient pretraining source for compact encoders, while complex text needs far more dimensions to expose useful structure.

1 Introduction

Sentence embeddings are mappings from high-dimensional linguistic signals to lower-dimensional structures whose geometry captures the semantic meanings of sentences. Latent space embeddings are crucial in NLP tasks. Recent progress—including BERT Devlin et al. [2019] for token embeddings and then Sentence-BERT Reimers and Gurevych [2019] for sentence embeddings—has significantly improved sentence embedding quality. In practice, resulting embeddings are typically used as is (in 768D or 1024D for BERT-based models) without any consideration given to decreasing the compute or memory on downstream tasks via a lower embedding dimensionality. However, the nature of the data itself and the downstream task might require a different dimensionality, which could make the resulting embeddings not only more efficient but also useful.

We train sentence representation models using a two-step procedure and evaluate their ability to generalize across domains. Our evaluation steps include 1) assessing transfer learning of the models trained on one particular dataset (either pop songs or scientific papers) on general, out-of-distribution tasks, and 2) the resulting geometry of in-domain and out-of-domain data embedded with our trained models. Through this analysis, we can also infer the intrinsic dimensionality of the topic-specific data used to train the models under evaluation.

On a high level, we take a step towards answering the following questions: Which conceptual domains (represented by the data used to train our models) allow for smaller-dimensional representations that still capture meaning and retain transfer learning abilities? How much can we decrease the embedding dimension while retaining useful representations?

2 Related Work

Widespread adoption of transformer encoders began with BERT Devlin et al. [2019], which learns deep contextual token representations that can be repurposed for sentence-level embeddings. Sentence-BERT Reimers and Gurevych [2019] extends BERT with a siamese architecture and pooling scheme, enabling efficient cosine-similarity retrieval and establishing a strong supervised benchmark for sentence embeddings. Li et al. [2020] analyze sentence embeddings produced by pretrained language models, showing strong anisotropy and proposing geometric corrections to improve isotropy and downstream task performance.

Contrastive learning provides the main algorithmic backbone for modern sentence encoders. Wang and Isola [2020] show that contrastive objectives implicitly trade off two geometric pressures—alignment of positive pairs and uniformity of embeddings on the hypersphere—offering a framework we directly use to interpret our models. Wang and Liu [2021] further study the behavior of contrastive losses such as InfoNCE under different design choices, explaining how temperature, sampling, and margin parameters induce distinct embedding geometries. Building on these ideas, SimCSE Gao et al. [2021] introduces a simple but powerful contrastive objective for sentences, where dropout-generated views serve as positive pairs and hard negatives emerge naturally from in-batch sampling.

Recent work also focuses on the dimensional structure of sentence representations. Wang et al. [2023] study the performance of low-dimensional sentence embeddings from pretrained encoders, decomposing the impact of the encoder and pooler and showing that substantial dimensionality reduction is often possible without dramatic loss in quality. Their work served as our main reference for this study. Amit et al. [2021] investigate how intrinsic dimensionality governs generalization, arguing that effective models adapt their representation dimensionality to the complexity of the underlying data manifold—a perspective that motivates our focus on how different corpora support useful low-dimensional sentence embeddings.

3 Methodology

We use contrastive learning as described in SimCSE Gao et al. [2021]. SimCSE’s training objective pulls positive pairs together in the embedding space and pushes negative pairs apart. In our context, positive and negative pairs are defined by closeness in the same document, creating a self-supervised setting. SimCSE was previously shown to be successful on STS benchmarks and transfer tasks, which is why we use it as the main method for learning sentence meaning. We train and evaluate two architectures: encoder + pooler (BERT + a feed-forward NN) and encoder + PCA (BERT + PCA). Below we briefly discuss implementation details. All the code is available at github.com/AlenaResiko/UML.

3.1 Encoder + Pooler

The encoder maps tokens to latent representations, and the pooler maps hidden states to low-dimensional embedding of sentences. Wang et al. [2023] found that performance loss can be decomposed into separate encoder and pooler contributions, which is the motivation for using a two-step training procedure. First, we freeze the parameters of the pooler and fine-tune the encoder, achieving the best version of BERT for a particular dataset. Then, we freeze the encoder and train only the pooler. We do each step in three epochs. This procedure allows us to reduce the encoder loss and the pooler loss sequentially.

We load `bert-base-uncased` weights from Hugging Face for BERT with the output in 768D. For the pooler, we construct a fully connected network with a hidden-size $\times$ hidden-size layer, tanh activation, followed by a hidden-size $\times$ projection-size fully-connected layer (following Wang et al. [2023]). We apply mean pooling to aggregate token embeddings into sentence embeddings.

The final output dimension is determined by the pooler. For our experiments, we sweep through $d \in \{4, 16, 64, 256, 767\}$.

Note that BERT produces position-aware token representations, so the order information describing tokens' relative positions in a sentence is not lost.

In training we apply the SimCSE trick: a sentence passed as is and then passed one more time with a dropout mask applied to the underlying model should have similar embeddings. For each anchor sentence, we compute the multi-positive normalized temperature-scaled cross-entropy loss and optimize using the Adam optimizer.

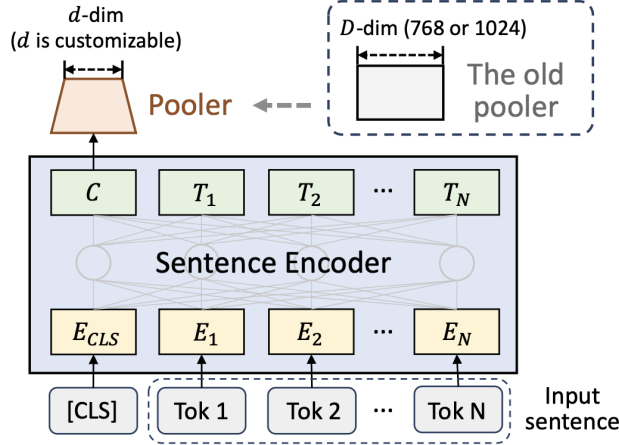


Figure 1: BERT+Pooler architecture. Image credit: <https://arxiv.org/pdf/2310.15285>.

3.2 Encoder + PCA

The encoder is again BERT, followed by PCA allowing us to customize the final output dimension. For a given projection dimensionality d , we take the first d principal components. We sweep through $d \in \{4, 16, 64, 256, 767\}$ to match the encoder + pooler experiments.

3.3 Training domains

We choose domains to capture different styles, word frequencies, and modes of communication:

- 1) Taylor Swift lyrics (short, repetitive, low-entropy)
- 2) Nakul Verma's ML papers (long, information-dense)

Each song and each paper section are treated as one document, and neighboring sentences within a document form positive pairs. The corpora are token-matched for training compute parity (since BERT operates on the token level, not sentence level). After pre-processing, we split the text into sentences, which we consider to be the smallest complete unit. Our pre-processing methodology is discussed below.

3.4 Data Pre-processing

For the lyrics dataset, we found a Kaggle repository containing the entire Taylor Swift discography and downloaded it for our lyrics corpus. The cleanup included removing sequential duplicate sentences, filtering out song and album metadata, and splitting the lyrics into individual sentences using the NLTK sentence tokenizer. Each newline was treated as an end-of-sentence marker.

For the scientific dataset, we used the arXiv API to download Nakul Verma's papers in pure LaTeX form. There was substantial cleaning and processing work involved from that point. First, we removed latex commands, equations, figures, tables, and references using regex patterns. Next, we split the cleaned text into sentences. These two actions required a fair amount of user judgement, and diminished the size of the usable corpus.

Given two corpora, we create pairs of similar sentences in a self-supervised manner through assigning similarity based on sentence distance: if sentences are within a distance 5 of each other, they are deemed similar.

3.5 Training Objective

The goal of contrastive learning is to pull together representations of positive examples and push apart negatives: in this way we learn correct neighborhood structure. We use contrastive training as in Gao et al. [2021]. Let P_i be a set of positive (near neighbors and itself) examples for each sentence and e_i be embeddings of each sentence. The per-pair similarity score function is a dot product normalized by temperature τ :

$$\exp S_{ij} = \exp\left(\frac{e_i^\top e_j}{\tau}\right)$$

We want to maximize the ratio of similarity of positives to similarity of non-positives, averaged over M pairs in a batch:

$$\mathcal{L} = \frac{1}{M} \sum_{i=1}^M -\log\left(\frac{\sum_{j \in P_i} \exp(S_{ij})}{\sum_{\substack{j=1 \\ j \neq i}}^M \exp(S_{ij})}\right).$$

Viewed as a probability, the function inside the log is a conditional probability $p(j|i)$ of positive examples j given an anchor sentence i . The training objective $\mathcal{L}$ is the cross-entropy over these conditional probabilities.

3.6 Evaluations

- Geometry:
 - Alignment and uniformity losses (Wang and Isola [2020]) on in-domain and out-of-domain test sets. Alignment measures how close positive pairs are, and uniformity captures how evenly embeddings are spread on the hypersphere (how much of the provided space they occupy).
 - t-SNE embeddings of sentences from test songs and papers. We aim to see if the two types are separated into distinct clusters by the models and examine the cluster geometry qualitatively.
- Transfer learning: STS-B Spearman rank correlations of sentence pair embeddings.

PCA projections of the frozen BERT encoder at the same target dimensions are used as a baseline.

4 Results and Analysis

4.1 Spearman Rank on STS-B

The results on lyrics-trained models rise sharply between 4D and 256D, peaking long before the full BERT size of 768D (see Fig. 2). Paper-trained models, in contrast, only come to full potential at 767D, hinting at a much **higher intrinsic dimensionality of scientific text**. Since STS-B is out-of-distribution for both training corpora, the gap between the performance of songs- and papers-trained models cannot be attributed to lyrics being much "closer" to STS-B than papers. In fact, qualitatively we could argue that STS-B is closer to scientific text than to pop song lyrics. Instead, the performance gap reflects how contrastive fine-tuning under a low-dimensional bottleneck **reallocates capacity of the model**.

On the lyrics corpus, the contrastive task is relatively simple and repetitive, so the model can solve it without drastically reshaping the pre-trained BERT geometry. A small output dimension is enough to compress lyrics-specific structure while preserving generic semantic directions that still transfer to STS-B.

On the papers corpus, the contrastive task is much harder: the model must separate dense, technical sentences within and across sections. In low dimensions, the encoder and the pooler are forced to specialize for fine-grained distinctions, which distorts general-purpose semantics and hurts STS-B

transfer. Only when we increase the projection size up to 256-767D does the paper-trained model regain enough capacity to encode both in-domain nuance and broad semantic structure, at which point it overtakes the lyrics-trained model.

The PCA baseline stays near zero Spearman rank across all projection sizes, suggesting that **linear compression alone cannot expose semantic structure** – the pooler must learn it.

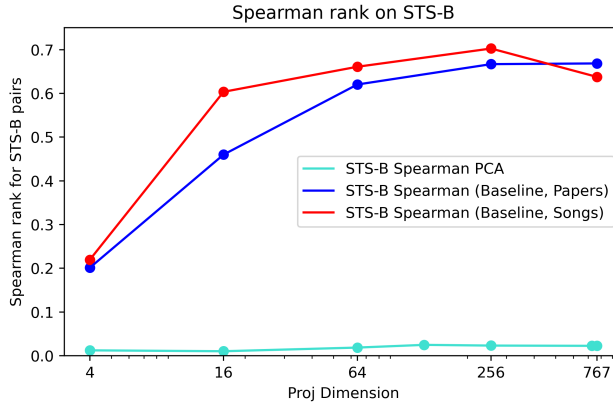


Figure 2: Spearman rank correlations for BERT+pooler and BERT+PCA variants.

4.2 Uniformity and Alignment

All plots in Fig. 3 show the **alignment vs uniformity tradeoff**: higher dimensions improve uniformity but degrade alignment. For PCA specifically, uniformity losses are much smaller than alignment losses for all dimensions, indicating that embedded sentences occupy the full hypersphere but do not preserve the local similarity structure.

For contrastive training, **in-domain data** (eg. songs embedded with a model trained on songs) **becomes well-spread** (good uniformity) but not overly tight, and **out-of-domain data** often collapses, achieving **artificially low alignment with poor global geometry**. This collapse likely occurs because the model is forced to project unfamiliar sentence structure into a space shaped only by in-domain positives, over-compressing local neighborhoods with unseen examples without preserving meaningful global structure. In addition, papers-trained models behave more similarly in terms of both losses on different domains, while songs-trained models have larger oscillations when domains are switched, hinting at more stable geometry learned by papers-based embeddings across domains.

4.3 Songs and Papers Clusters Embedded with t-SNE

Even with only 4 dimensions, the embeddings already separate lyrics from papers as two distinct clusters (see Fig. 4), indicating that coarse domain identity survives aggressive compression. However, in 4D the clusters are elongated and less compact, with visibly distorted shapes. In addition, there is a group of paper sentences that end up being much closer to the songs cluster (concentrated in the bottom right part of the plot). At 16 dimensions, the clusters become rounder, and local neighborhoods look more stable. This suggests that increasing dimensionality from 4D to 16D does not create domain separation from scratch, but instead improves the within-domain geometry: it reduces compression artifacts and yields a smoother manifold, which is consistent with the concurrent improvement in STS-B performance.

4.4 Ablation on the Training Procedure

For songs-trained models, with smaller dimensions (4, 16), the two-step training process yields the best Spearman rank correlations, showing how **important the pooler is for extreme dimensionality reduction** (see Fig. 5). However, at certain higher dimensions training only the encoder yields the best results. Training only the pooler is not sufficient for learning useful representations for songs-based models.

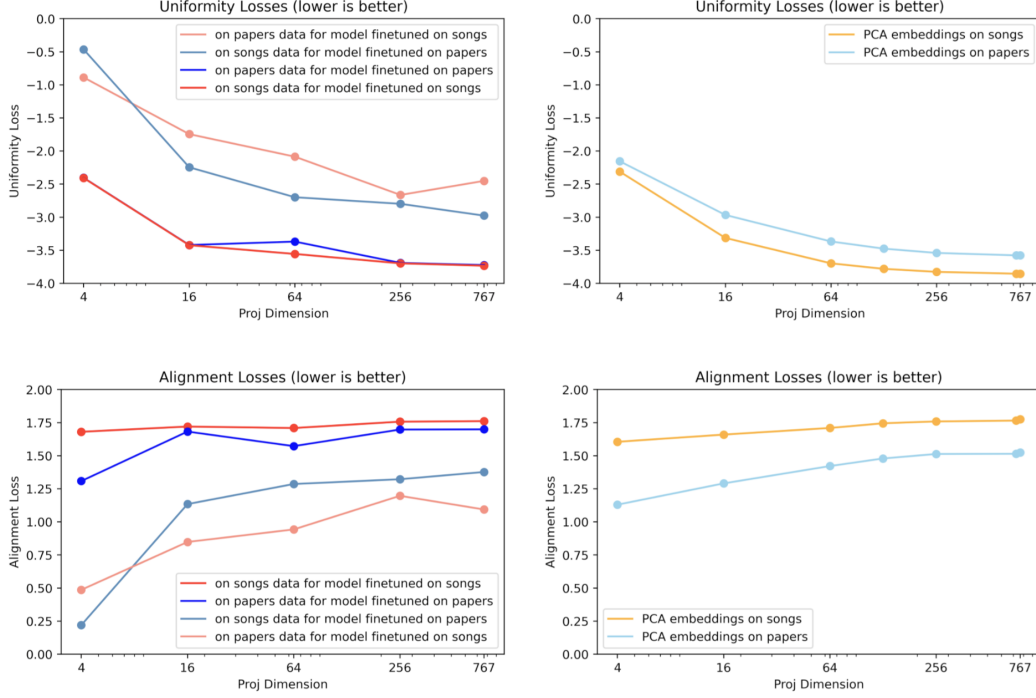


Figure 3: Uniformity and alignment losses.

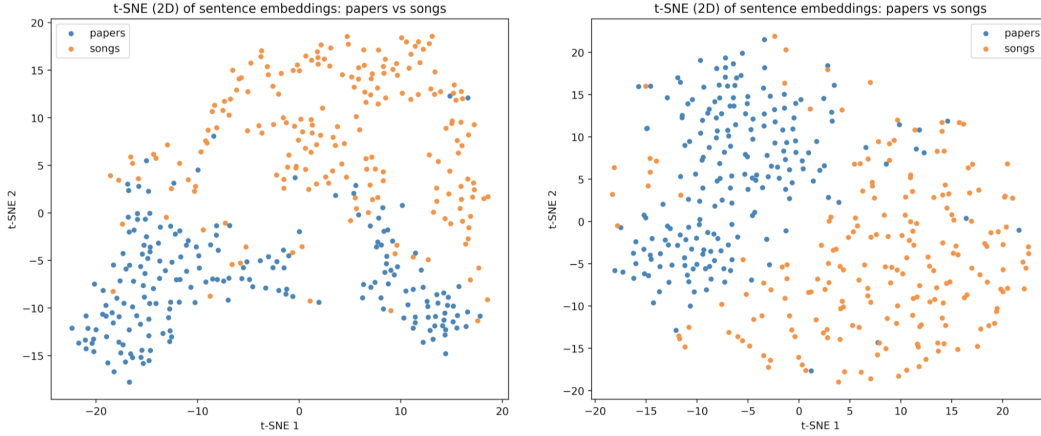


Figure 4: t-SNE results for 4D and 16D BERT+pooler models trained on songs.

For papers-based models, we see a similar trend, but the dimension cutoff at which the pooler stops giving much benefit is higher, somewhere between 128D and 256D (again hinting at the higher intrinsic dimensionality of scientific data). In addition, only training the pooler with the frozen BERT backbone yields much better Spearman rank results for papers-trained models than for songs-trained models, perhaps hinting that the frozen BERT encoder is initially better suited for academic text.

5 Conclusion

5.1 Summary of Findings

- **Lyrics-trained models dominate at low dimensions (4–256D) on STS-B**, while papers-trained models only overtake at 767D $\Rightarrow$ **scientific text appears inherently higher-dimensional**.

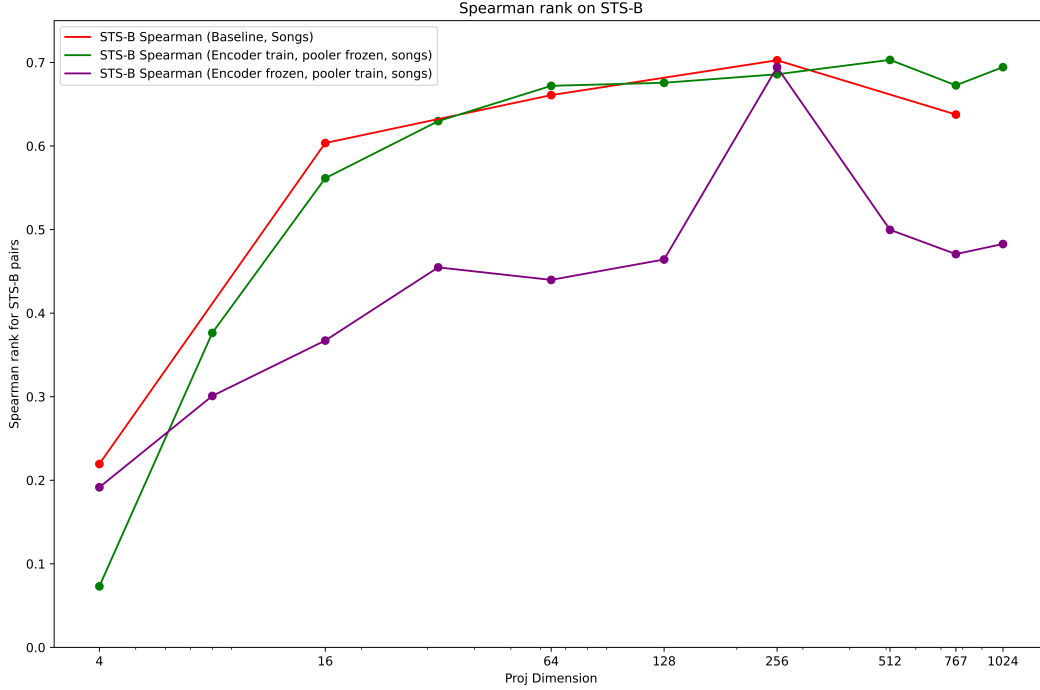


Figure 5: Ablation study on the two-step training procedure.

- **Uniformity and alignment conflict** with each other, with **papers-trained models being more stable** when transitioning from in-distribution to out-of-distribution data.
- **Low-dimensional embeddings** are able to **separate domains (even at 4D)**, but more meaningful local structure emerges only at 16D or more.
- Linear compression (**PCA**) is **insufficient for transfer learning** tasks in our experiments: contrastive learning is necessary to expose semantic structure even at small dimensions. We recommend retraining, rather than linearly transforming embeddings, when the target dimensionality changes.
- The **two-step training** procedure of the encoder + pooler setup is **necessary** for satisfactory performance in **extreme dimensionality reduction** settings. When we increase the output dimension, training the pooler becomes less important.

5.2 Challenges and Limitations

We note that all reported results depend on a specific choice of backbone (BERT-base), corpus size, and contrastive neighborhood definition. In addition, while STS-B Spearman rank results provide an insightful semantic probe, it represents only one notion of transfer learning performance.

Defining Similarity Drawing inspiration from Gao et al. [2021], we explored enriching our unlabeled dataset with self-supervised positive pairs. We landed on sentence proximity as a straightforward and effective method to generate these pairs without additional labeling. However, many other approaches exist: only using the same sentence embedded by the same network with random dropout masks twice is one interesting example. One could also use a large network to label similarity, allowing for knowledge distillation. Our approach could deem unrelated sentences that appear consecutively in a document similar, which is an important limitation introducing a lot of noise.

Contrastive Loss Fine-tuning By design, contrastive fine-tuning mainly acts on local geometry: it pulls positive pairs together and pushes negatives apart, encouraging a more uniform distribution of representations on the unit hypersphere. Because most of the representational structure is inherited from the pretrained BERT encoder, the optimization is effectively upper-bounded by these initial

representations. Our procedure does not attempt, nor is it capable of, substantially re-architecting the global geometry of the embedding space. Thus, while contrastive learning can refine local neighborhoods and yield meaningful improvements, it is not expected to produce dramatic, paradigm-shifting gains in sentence embedding quality.

Scope Due to time and resource constraints, we limited our experiments to two narrow domains of the English language. A more comprehensive study would include a much broader array of domains, self-labeling algorithms, and backbone architectures beyond BERT-base-uncased (e.g., RoBERTa, DeBERTa, etc.).

5.3 Future Directions

For future work, we would like to explore **additional domains** (such as news, code comments, etc.), try other **embedding models beyond BERT**, use **alternative training objectives** (for instance, non-contrastive methods), enrich the suite of **evaluation methods**, and implement competing self-supervised **sentence pair labeling algorithms**. This wider range of methods can allow for better capture of sentence structure’s geometry.

Expanding the scope a bit, it would be interesting to move beyond all-purpose sentence embeddings and explore **task-specific embeddings** and **multimodal embeddings**, modeling language in a **hierarchical manner**, moving beyond Euclidean spaces into **non-Euclidean geometries**, and exploring the notion of **intrinsic dimensionality** more rigorously.

5.4 Contributions of Each Team Member

- **A.C.:** Partial preprocessing, BERT+pooler contrast-and-freeze architecture implementation, BERT+PCA architecture implementation and evaluation on 4-767D, partial report write-up.
- **A.K-W.:** Data collection and preprocessing for both datasets, ablation study on the BERT+pooler architecture for both datasets with training and evaluation on 4-1024D, partial report write-up.
- **M.G.:** Initial literature review, evaluation suite and graphics implementation, baseline BERT+pooler training and evaluation on 4-767D for both datasets, poster preparation, partial report write-up.

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Code Availability

All our code can be found at github.com/AlenaResiko/UML.

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